UKA TARSADIA UNIVERSITY

Integrated M.Sc. (IT) (Semester 4)
060010409(2016-17)
CC11 Java Programming

Date :20/04/2019 **Time :**9:30AM- 12:30PM
Max. Marks:60

Instructions :

1. Attempt all questions.
2. Write each section in a separate answer book.
3. Make suitable assumptions wherever necessary.
4. Draw diagrams/figures whenever necessary.
5. Figures to the right indicate full marks allocated to that question.
6. Follow usual meaning of notations/abbreviations.

SECTION - 1

Q 1 A) Answer the following. **[4]**

I) What is finalize() method?

II) State the importance of final keyword.

III) Write a usage of escape sequence.

IV) Define: Identifiers.

Q 1 B) Answer the following in brief. (Any 3) **[6]**

I) Why Java called the “Platform independent programming language”?

II) What is jagged array? How one can define it?

III) Enlist two type of data type conversion. Give an example for same.

IV) State and define two differences between static method and final method.

Q 2 Answer the following. **[10]**

A) In a class, one method has two overloaded forms. One form is defined as static and another form is defined as non-static. Is that method properly overloaded? Refer following code and identify the error and give reason for same.

```
class Hello {  int x,y;
        Hello() { x=y=10; }

        public int display(int a){
                x=a;
                System.out.println("Val " + x);
                return x; }
        private float display(int b) {
                y=b;
                System.out.println("val " + y);
                return y; }
        protected void display() {
                System.out.println("Val is " + x + " " + y);
                this();   }

        public static void main(String args[]){
                Hello h=new Hello();
                h.display(20);
                h.display(); } }
```

OR

A) What do you mean by interface? Can we declare final to it? How does it differ from abstract class? Give example of it.

B) An “Aaradhya bank” situated in Bhuj having 100 accounts in their bank. Generally a bank account contains data member as account number, name of customer, type of account and amount. Design a class to model the bank account. Write a menu driven program to perform basic operations on bank account like deposit, withdraw and display account details.
(Note: Create parameterized constructor to initialize bank accounts information and create a member method as depositeAmount, withdrawAmount and displayAccountInfo to perform above functionalities.)

OR

B) Create one Person class which have data member like personName, height and weight and method is display(). Faculty class inherit the properties of Person and it has data member like degree and method is display(). Create an object of base class and pass reference of derived class. Now call display() method with that base class object. Which display() method will call? Justify your statement.

Q 3 Answer the following in detail. (Any 2) **[10]**

I) What is an inheritance? Discussed advantage and use of inheritance. Provide a sample code to represent member accessibility in inheritance.

II) Which method is used to create a directory in Java? Write a snippet code for creating directory and subdirectory in it. Explain how to create and access package in Java with an example.

III) Write any four feature of an array. Consider a scenario where user wants to create one integer type array which has three rows and having three, one and two column respectively. Draw a structure for current scenario and write a program to create specified array, take element input from user and display all elements of an array.

SECTION - 2

Q 4 A) Answer the following. **[4]**

I) Can we call String class method using String literals? If yes then give one example for same.

II) Refer the below code, if there is an error solve it and display output for the same. . (Assume that code is written main method of Exam class.)

```
try{
System.out.println("In try block"); }
finally{
System.out.println("In finally block");}
```

III) Write one key difference between read() and readLine().

IV) Write one difference between Set and List.

Q 4 B) Answer the following in brief. (Any 3) **[6]**

I) How to convert integer to String and String to integer in Java? Give one example for same.

II) List and define two ways to iterate the element of collection.

III) How one can join three strings without using + operator? Give one example of it.

IV) What is finally block? Can finally block is used without catch? If yes then give example.

Q 5 Answer the following. **[10]**

A) Create an application which accept password that should have minimum six characters.When user enter less than six characters in a password throw custom exception. Write a program which handle the exception whenever it raise.

OR

A) Write a program which will take input of personal information of person from console and store it into person.txt file.

B) Write a program which will display total number of words from the person.txt file.

OR

B) Write a program which will insert new employee details into Employee(eid int, name varchar(30),address varchar(50),contactno int) table.

Q 6 Answer the following in detail. (Any 2) **[10]**

I) What is JDBC API? Discuss various components of it.

II) What is the role of JDBC driver? Explain any two types of JDBC driver with its advantages and disadvantages.

III) Explain StringTokenizer class with an example. State any two differences between String class and StringTokenizer class.

UKA TARSADIA UNIVERSITY

Integrated M.Sc. (IT) (Semester 4)
060010409(2016-17)
CC11 Java Programming

Date :15/09/2018 **Time :**9:30AM- 12:30PM
Max. Marks:60

Instructions :

1. Attempt all questions.
2. Write each section in a separate answer book.
3. Make suitable assumptions wherever necessary.
4. Draw diagrams/figures whenever necessary.
5. Figures to the right indicate full marks allocated to that question.
6. Follow usual meaning of notations/abbreviations.

SECTION - 1

Q 1 A) Answer the following. **[4]**

I) How one can initialize blank final variable?

II) State the importance of byte code.

III) What is the reason of declaring main method as static in Java?

IV) Define: Boxing Conversion

Q 1 B) Answer the following in brief. (Any 3) **[6]**

I) Briefly define execution process of a simple Java program.

II) Provide an output of following,

a. 15 ^ 10
b. 9 << 2
c. ~ 12
d. -1 >>> 24

III) What is character set? Which character set does Java use?

IV) What will be the output of below code? (Assume that there is no syntax error in code and code written in Demo class which has main method)

```
System.out.printf("%d %c %f \n", 1,'A',10.10);
System.out.printf("Number is %05d", 123);
System.out.printf("Number is %.4f", 12.987654);

String myString = "Hello friend,\n\tI am from \"M.Sc.(IT)\"";
System.out.println(myString);

float a = 100.551f;
byte c=67;
int b = (int)a;
char d=(char)c;
System.out.println("Value of b is : " + b + "\n Value of d is : " + d);
```

Q 2 Answer the following. **[10]**

A) What is the importance of nested class in Java? Illustrate the types of nested class with an example.

OR

A) When we need to use extends and implements? Give on example for same. Refer following code and identify the error and give reason for same.

```
class A {
    int a;
    A(){ a=10; }

    public void display() {
        a=20;   System.out.println(a); }

    final void show(){
        a=30;   System.out.println(a); } }

class B extends A {
    int b;
    public void display() {
        b=40;   System.out.println(b);          super(); }


     void show(){
        a=40;   System.out.println(a); } }

class Dynamic_dispatch {
    public static void main(String args[]){
            A obj1 = new A();
            B obj2 = new B();
            A obj3=obj1;
            obj3.display();
            obj3 = obj2;
            obj3.display();
            B obj4=obj1;
    obj4.display(); } }
```

B) Write a program for “George Company Pvt. Ltd.”, who wants to manage the employee information like name, address, hourlyWages, month and year of joining, etc.). Model a class which will maintain data of 100 employees. Display the detail of employee who has been with the company 20 years or more.

OR

B) State difference between static binding and dynamic binding. Give output of following segment of code,

```
class Home{
    public void squareFeet(){
        System.out.println("Squarefeet depends upon Lower, Medium and Higher income
    group"); } }

class LIG extends Home{
    public void squareFeet(){
         System.out.println("Lower Income Group: SQ is 3 cent"); } }

class MIG extends Home{
    public void squareFeet(){
        System.out.println("Medium Income Group: SQ is 5 cent"); } }

class HIG extends Home{
    public void squareFeet(){
        System.out.println("Higher Income Group: SQ is 8 cent"); } }
```

```
class Demo{
    public static void main(String args[]){
    Home home=new Home();
    LIG l=new LIG(); MIG m=new MIG(); HIG h=new HIG();
    Home home1=l; Home home2=m; Home home3=h;

    home.squareFeet(); l.squareFeet(); m.squareFeet();  h.squareFeet();
    home1.squareFeet(); home2.squareFeet(); home3.squareFeet(); } }
```

Q 3 Answer the following in detail. (Any 2) **[10]**

I) What are abstract classes? Differentiate abstract class with final class.

II) Why switch case is better than multiple if else statement? Demonstrate with an example.

III) What is inheritance? Among the following explain any two in detail.

i) Super keyword
ii) Types of inheritance
iii) Advantages of inheritance

SECTION - 2

Q 4 A) Answer the following. **[4]**

I) What is Exception?

II) Define collection framework.

III) Which exception is thrown by given code?

```
void compute()
{       int a= 19;        int b = 0;        System.out.println("Result :-"+a/b);    }
```

IV) What is default delimiter in StringTokenizer?

Q 4 B) Answer the following in brief. (Any 3) **[6]**

I) Give one key difference between Abstract class and Interface. Justify your answer with appropriate example.

II) Write two benefits of an exception handler.

III) How to create String from character array? Give one example.

IV) What is the output of following code? How many String objects and how many reference variables were created prior to the println statement? (Assume the given code is written in main method of Exam class.)

```
String s1 = "spring ";
String s2 = s1 + "summer ";
s1.concat("fall ");
s2.concat(s1);
s1 += "winter ";
System.out.println(s1 + " " + s2);
```

Q 5 Answer the following. **[10]**

A) Which classes are used to read and write bytes from and into a file? Write a Java program to fulfill following requirement,

1. Take character input from user using character stream.
2. Store the byte value of that character into "Demo.txt" file.
3. Count total number of bytes in a given file.

OR

A) Write a program for MovieTicketBooking, which will take movie ticket booking information from user and store it into TicketBooking table using PreparedStatement.
[Note: Assume table is already created in MYSQL, fields for TicketBooking table are – TicketID, MovieName, DateOfShow, TimeOfShow, BasicPrice, NumberOfPerson, Amount]

B) Write a program which will take employee information like, empId, empName, empAddress and empDesignation using character stream. Throw an user define exception whenever user enter wrong empId. empId must be starts with company name like SUPERMART101, where SUPERMART is company name and 101 is unique id of employee.

OR

B) Draft a procedure as "UpdatePersonDetail" in MYSQL to update PersonName, ContactNumber based on AadharNo. Write a Java program to execute that procedure. [Assume Person table is already created in MYSQL as Person (AadharNo, PersonName, Address, Gender, ContactNumber).]

Q 6 Answer the following in detail. (Any 2) [10]

I) Explain StringTokenizer class with its overloaded form of constructor and any three methods.

II) Write a detail note on JDBC drivers.

III) Discuss 5 steps to connect the database in Java.

UKA TARSADIA UNIVERSITY

Integrated M.Sc. (IT) (Semester 4)
060010409(2016-17)
CC11 Java Programming

Date :08/05/2017 **Time :**9:30AM- 12:30PM
Max. Marks:60

Instructions :

1. Attempt all questions.
2. Write each section in a separate answer book.
3. Make suitable assumptions wherever necessary.
4. Draw diagrams/figures whenever necessary.
5. Figures to the right indicate full marks allocated to that question.
6. Follow usual meaning of notations/abbreviations.

SECTION - 1

Q 1 A) Answer the following. **[4]**

I) How to identify the size of column in ragged array? GIve one example.

II) Write one usage of block statement.

III) In a class, what is the default scope of the variable and what is the default access modifier of it?

IV) Define class.

Q 1 B) Answer the following in brief. (Any 3) **[6]**

I) List the features of Java Language and write any two features of it.

II) Define significance of “public static void main(String args[])”.

III) Refer the below code and display output. (Assume that there is no syntax error in code.)

```
public class Exam
{  public static void main(String args[]) {
int i,j,k,l=0;
k=l++;
j=++k;
i=j++;
System.out.println(i);
}}
```

IV) What are escape sequence and why they are needed?

Q 2 Answer the following. **[10]**

A) Do as directed.

```
class Student{
   int id;  String name;

   Student(){
       id=0; name=null;
       System.out.println("In default Constructor \n"+id+" "+name);}

   Student(int i,String n){
     id = i; name = n;
     System.out.println("In parametrized Constructor \n"+id+" "+name);}

  Student(Student s){
     id = s.id; name =s.name;
     System.out.println("In copy Constructor \n"+id+" "+name);}
```

```
public static void main(String args[]){
Student s1 = new Student(101,"Manav");
Student s2 = new Student(s1);}}
```

i) State two difference between Java and C++.

ii) Which keyword is used to call default constructor?

iii) What will be the output of above given code after answering (ii) question and putting that keyword on first statement of parameterized constructor?

OR

A) A Classroom having number of students and benches. By consider a scenario of Classroom use the concept of nested class by which user can initialize and view the detail of students in particular classroom as well as number of benches available in classroom.
(Note: Use following member variable to design above functionality.
Classroom – Room_Number
Student – stud_Id, stud_Name
Bench – row, column)

B) Can we override final method in Java? What are the difference between method overloading and method overriding? Illustrate with proper example.

OR

B) Create an abstract class Accounts with the following details:
Data members: Balance, accountNumber, accountHolderName, address
Methods: withdraw(), deposit(), display()
Withdraw and deposit should be an abstract method
Create a subclass as SavingAccount and CurrentAccount with following details:
For SavingAccount use,
Data members: rateOfInterest,
Methods: calculateInerest(), display() - to display rate of interest with new balance and full account holder details.
For CurrentAccount use,
Data members: overDraftLimit
Method: display() to show overdraft limit along with the full account holder details.
Write a program to open an account of user choice and display its information in proper manner.

Q 3 Answer the following in detail. (Any 2) **[10]**

I) Define: Inheritance. Explain need of inheritance with suitable example.

II) What is the difference between regular array and jagged array? Demonstrate jagged array in detail.

III) What is abstract method? Can you use abstract and final both with a method? Explain the importance of abstarct class with suitable example.

SECTION - 2

Q 4 A) Answer the following. **[4]**

I) How many object will be created in the following code and where they will be stored in the memory?
String str1="BMIIT";
String str2="BMIIT";

II) Which keyword is used to monitor statement for exception?

III) State one key significance of StringTokenizer class.

IV) Write one key difference between final and finally.

Q 4 B) Answer the following in brief. (Any 3) **[6]**

I) Give one key difference between String class and StringBuffer class. Justify your answer with appropriate example.

II) Write any two benefits of collection framework.

III) Write two key usage of try and catch.

IV) Refer the below code and give one key difference between length() and capacity(). Write output for the same. (Assume that the given code is written in main method of Exam class.)
StringBuffer sb=new StringBuffer("Hello");
System.out.println("sb.length() = " + sb.length());
System.out.println("sb.capacity() = " + sb.capacity());

Q 5 Answer the following. **[10]**

A) Write a program which will increase the salary of employee who are working in marketing department using PreparedStatement. Assume that in database there is a table Employee with the fields id,name,department and salary.

OR

A) Write a Java Program which will read a string and rewrite it in the alphabetical order. For example: The word “STRING” should be written a “GINRST”.

B) Write a java program which will insert one line through stream class and count total no of a repeat word from the line.

OR

B) Write a program which will throw an exception whenever user does not enter his/her name in capital letter.

Q 6 Answer the following in detail. (Any 2) **[10]**

I) List and explain four types of JDBC driver with its advantages and disadvantages.

II) State the significance of StringTokenizer class. Discuss three constructor of StringTokkenizer class.

III) How to connect database and perform operation in java? Explain with an example.

UKA TARSADIA UNIVERSITY

B.Sc. (IT)/Integrated M.Sc. (IT) (Semester 4)
IT4008(2020-21)
Java Programming

Date :28/11/2022 **Time :**9:30AM- 12:30PM
Max. Marks:60

Instructions :

1. Attempt all questions.
2. Write each section in a separate answer book.
3. Make suitable assumptions wherever necessary.
4. Draw diagrams/figures whenever necessary.
5. Figures to the right indicate full marks allocated to that question.
6. Follow usual meaning of notations/abbreviations.

SECTION - 1

Q 1 A) Answer the following. **[4]**

I) What is the role of JIT in JRE?

II) What is the difference between constant and variable in Java?

III) Why Java programs are able to run on different operating system?

IV) Can a multidimensional array in Java be uneven (Jagged)?

Q 1 B) Answer the following in brief. (Any 3) **[6]**

I) Below code generates compilation error. State True/ False with proper justification

```
class Example {

public static void Main(String s[]) {
System.out.print ("This is a simple Java program.") ;}
}
```

II) Consider the below java code

```
class UT1{
   String FirstName, LastName;
   public void joinName (String FirstName){ System.out.println(FirstName);}
   public void joinName (String FirstName, String LastName){
      System.out.println(FirstName +" "+LastName);
   }
   public static void main(String[] arg){
      ____________________________________
      u.joinName("Niyati","D.","Raval");
   }
}
```

Answer the below 2 questions (I, II) on the basis of above code.
1. What changes should be made in above code so that the u.joinName(“Niyati”,”D.”,”Raval”); statement will give the output as follow:
   Niyati D. Raval.
2. Write a code for blank box which execute this above code without any error.

III) State any four features of Static Field and Method.

IV) What do you mean by implicit and explicit type conversion? Explain with suitable example.

Q 2 Answer the following. [10]

A) What would be the output of following code? Do sample calculation.

```
class MyClass{
public static void main(String [] args){
        boolean b1=false, b2=false;
        int x = 0;
        if ( !b1 ){
                if ( !b2 ){
                                b1 = true; x++;
                                        if ( 9 > 6 )
                                                x++;
                                                if ( !b1 )
                                                x = x + 20;
                                                else if ( b2 = true )
                                                x = x + 220;
                                                else if ( b1 | b2 )
                                                x = x + 1000;
                        }
                }
        System.out.println(x);
        }
}
```

OR

A) Refer following code and identify the error and give the reason for same.

```
1)
  boolean b=true;
  int x=b;
 System.out.println("Value of x is : " + x);
2)
 class Student{
        int sid;
        String sname;
        Student(){
        sid=0;
        sname=null;
       }
        Student(int id, String name){
        sid=id;
        sname=sid;
        this();
        }
}
```

B) A book shop maintains the inventory of books that are being sold at the shop. The list includes details such as author, title, publisher, cost, and stock. Whenever a customer wants a book, the sales person inputs the title and author and the system searches the list and displays whether it is available or not. If it is not, an appropriate message is displayed. If it is, then the system displays the book details and requests for the number of copies required. If the required copies are available, the total cost of the requested copies is displayed, otherwise, the message "required copies not in stock" is displayed.

Design the classes to implement above given problem definition and demonstrate it.

OR

B) Design a class to represent a bank account. Include the following members:
Data members

- Name of the depositor
- Account number
- Type of account
- Balance amount in the account

Methods

- To assign initial values
- To deposit an amount
- To withdraw an amount after checking balance
- To display the name and balance

Design appropriate constructor to provide initial values.

Q 3 Answer the following in detail. (Any 2) **[10]**

I) Do as directed:
1. Describe the switch statement.
2. Write a syntax of switch statement.
3. What happens when we forget to put a break statement in a case clause of a switch?
4. List decision making statements of Java.
5. Differentiate break and continue.

II) Define array. How to initialize an array of float with 15 size? Write a snippet code to search specific element from an array.

III) What is package? Which package is always imported by default? How to create user defined package in Java. Explain it using appropriate example.

SECTION - 2

Q 4 A) Answer the following. **[4]**

I) State the key difference between concat() and join() of String class.

II) Write the purpose of writing finally block in exception handling.

III) Define: Set Collection Framework.

IV) Write the usage of StringBuffer class.

Q 4 B) Answer the following in brief. (Any 3) **[6]**

I) Analyse the given code and write what will happen when you compile and execute below program. [If program raise error justify reason for same else write output]

```
class SampleException
{
    public static void main(String arg[])
    {
        int a,b=0;
        try { a=50/b;  }
        catch (Exception e)
        {
            System.out.println("General Exception");
        }
        catch (ArithmeticException e)
        {
            System.out.println("ArithmeticException");
        }
        System.out.println("After catch statement");
    }
}
```

II) Justify – “String are immutable".

III) Write the output and justification for variable ans in below program :

```
Class Demo
{
    Public static void main(String args[])
    {
        boolean ans;
        ans=("java".toUpperCase().equals("java"));
        System.out.println(ans);
        ans=("java".toUpperCase()==("java"));
        System.out.println(ans);
    }
}
```

IV) Narrate any four differences with example and syntax for throw and throws.

Q 5 Answer the following. **[10]**

A) Write a difference between execute( ), executeQuery( ), executeUpdate( ) with example.

OR

A) List and discuss the steps to connect database with Java.

B) Consider a table called Dominos(pizzaname, size, price) which is already created. Write a program to place data of pizza's whose size is medium in file "Medium.txt". Stores data to file in given format only,

Format : Name|Size|Price
Example:
Cheese|M|250
Mexican|M|320

OR

B) Write a program which will accept a string and rewrite it with moving all upper case characters to the end of string in the following order.
Input: UkaTarsadiaUniversity
Output: kaarasadianiversityUTU

Q 6 Answer the following in detail. (Any 2) **[10]**

I) Discuss various methods and constructor of stringTokenizer class with an example.

II) Explain JDBC drivers with its advantages and disadvantages.

III) Explain steps required to established JDBC connectivity in Java.

UKA TARSADIA UNIVERSITY

B.Sc. (IT)/Integrated M.Sc. (IT) (Semester 4)

IT4019(2022-23)

Essentials of Advanced Object-Oriented Programming

Date :21/04/2023 **Time :**9:30AM- 12:30PM

Max. Marks:60

Instructions :

1. Attempt all questions.
2. Write each section in a separate answer book.
3. Make suitable assumptions wherever necessary.
4. Draw diagrams/figures whenever necessary.
5. Figures to the right indicate full marks allocated to that question.
6. Follow usual meaning of notations/abbreviations.

SECTION - 1

Q 1 A) Answer the following. **[4]**

I) How can you modify a string in Java without violating its immutability?

II) Define: Exception

III) Why string class is final in Java?

IV) What is the default capacity of StringBuffer class?

Q 1 B) Answer the following in brief. (Any 3) **[6]**

I) Draw the hierarchy of Built-in exceptions in java.

II) Give the output of following code and justify your answer.

```
String s = “hello”;
s.concat(“world”);
System.out.println(s);
```

III) Write the output of following code snippet.

```
String str3 = "Hello World";
String result1 = str3.substring(6, 11);
System.out.println(result1);
```

IV) What is the output of following code snippet.

```
String str7 = "The quick brown fox jumps over the lazy dog";
StringTokenizer st1 = new StringTokenizer(str7);
while(st1.hasMoreTokens()) {
        System.out.println(st1.nextToken());
}
```

Q 2 Answer the following. **[10]**

A) Write code to accoplish the following tasks:
Create a class called shape with the attribute X and Y and method calculateArea().
Create an interface called Drawable, which contains the draw() method.
Also the Derive class rectangle with the attributes like width and height and method calculatePerimeter().
Derive class circle with the attribute like radius and method calculateCircumference().
Create a controlling class and implement appropriate relationship between the classes and interfaces.
draw() method return the shape.

OR

A) Create an interface fruit which contains makeJuice() method. Classes Banana and Mango implements the fruit interface and override the makejuice() method. Create the controlling class and implement appropriate relationship between the classes and interfaces. makejuice() method will return the fruit juice.

B) Write a program that uses a Deque interface to implement a buffer. Your program should allow users to add and remove elements from the buffer, and to retrieve elements from both ends of the deque.

OR

B) Create a linked list of doubles and add five different values to it. Then, calculate the average of the values in the list and print out the result.

Q 3 Answer the following in detail. (Any 2) **[10]**

I) What role do interfaces play in java? Write the interface declaration with all the possible type of methods and variable along with their definition in other class.

II) Why exception handling is needed? Explain class hierarchy to explain different exceptinos.

III) Write a program to create user defined exception to handle the situation where withdrawal balance is more than available balance. Provide both the values as a parameter to validate() method.
Note: Display the tracing of the exception.

SECTION - 2

Q 4 A) Answer the following. **[4]**

I) Draw the format of UDP packet.

II) What is the purpose of using the parameter 'N' in generic programming?

III) State the usage of readUTF().

IV) What is the usage of receive() method in DatagramSocket?

Q 4 B) Answer the following in brief. (Any 3) **[6]**

I) Write the usage of parameter ‘S’, ‘U’, ‘V’ in generic programming.

II) Write a code to create packet for UDP socket.

III) Write a code to assign socket input stream to InputStreamReader?

IV) Write syntax to define generic constructor.

Q 5 Answer the following. **[10]**

A) Create a program that creates three threads by extending the Thread class. Each thread should print out the first 10 prime numbers.

OR

A) Write a Java program to implement a deadlock situation using threads. The program should create two threads, each of which needs a resource held by the other thread, resulting in a deadlock.

B) Create a program that implements a simple chat application using UDP datagrams in Java. Allow multiple clients to send and receive messages in real-time.

OR

B) Write a program to create server and client for service-oriented computing. Server is calculating square of the number provided by the client. Make sure that it only accepts number.
Use TCP/IP connection to implement such functionality.

Q 6 Answer the following in detail. (Any 2) **[10]**

I) What is upper-bound wildcard? Explain the benefits of using the upper-bound wildcard by giving example.

II) Discuss multithreading in detail.

III) State benefits of using generic programming and discuss the restrictions enforced on generic programming.

UKA TARSADIA UNIVERSITY

B.Sc. (IT)/Integrated M.Sc. (IT) (Semester 4)
IT4008(2020-21)
Java Programming

Date :17/04/2023 **Time :**9:30AM- 12:30PM
Max. Marks:60

Instructions :

1. Attempt all questions.
2. Write each section in a separate answer book.
3. Make suitable assumptions wherever necessary.
4. Draw diagrams/figures whenever necessary.
5. Figures to the right indicate full marks allocated to that question.
6. Follow usual meaning of notations/abbreviations.

SECTION - 1

Q 1 A) Answer the following. **[4]**

I) In what situation "this" keyword should be used?

II) What is JIT?

III) How JVM provides sendboxing?

IV) Enlist all the escape sequences.

Q 1 B) Answer the following in brief. (Any 3) **[6]**

I) Write two line code for garbage collection.

II) Write a code to demonstrate command line argument. (Write only required lines)

III) Provide example of narrowing and widening type conversion.

IV) Write spacial characteirstics of static method and field.

Q 2 Answer the following. **[10]**

A) Consider an example of declaring the examination result. Design three classes: Student, Exam and Result. The Student class has data members such as those representing roll number, name, etc. Create the class Exam by inheriting Student class. The Exam class adds fields representing the marks scored in six subjects. Derive the Result from the Exam class, and it has its own fields such as total_marks. Write an interactive program to model this relationship to produce result.

OR

A) How polymorphism takes place in interfaces? Discuss with an example.

B) Create two class one is Personal and another is Professional, Personal class has data member like name, address and contact_no, Professional class inherit the properties of Personal class and it has data member like company_name , designation, and salary and do the following operation.
1. Create two constructors for base and derived class.
2. Insert data through parameterized constructor of derived class.
3. Display all the information through parent class reference variable.

OR

B) Differentiate method overloading and method overriding with a suitable example. Discuss the possible way to access final variable in derived class.

Q 3 Answer the following in detail. (Any 2) **[10]**

I) How to create a two-dimensional array? Write a program to create ragged array, insert and display data to/from ragged array. [Note: Row size and number of columns per row size will be given by user.]

II) What is an inheritance? Discuss advantage and use of inheritance. Provide a sample code to represent member accessibility in inheritance.

III) Demonstrate the usage of Java package. Discuss various level of access protection for package and their implication.

SECTION - 2

Q 4 A) Answer the following. **[4]**

I) Write a code to raise NUllPointerException.

II) Which keyword is used for user defined exception?

III) State the usage of trim() method.

IV) What is the special characteristic of finally block?

Q 4 B) Answer the following in brief. (Any 3) **[6]**

I) String str="Hello";
String str1="Hello";
String str2=new String("Welcome");
String str3=new String("Welcome");
How many total number of objects will be created upon executing these statements?

II) Determine the category that the valueOf() method belongs to amongst the provided below.
(Abstract method, Instance method, Static method or Default method)
Enlist the datatypes not supported by valueOf() method.

III) Draw exception handling flow diagram.

IV) How StringBuffer class is different than String class?

Q 5 Answer the following. **[10]**

A) Consider table :
Tbl_employee_master (Employeeid,Employee_name,Employee_Email,Employee_Password, Employee_status)

Take employee name and employee password from user and check user is allowed to access system or not. If he/she is valid user then display welcome message and current system date and time along with inserted email address.

Note: Password is stored using MD5 encrypted algorithm.
Example
Email : RishiGupta@gmail.com password : abcde
Output: Welcome RishiGupta@gmail.com

OR

A) Consider table :
Tbl_employee_master (Employeeid,Employee_name,Employee_Email,Employee_Password, Employee_status)

Take employee email id from user and check that record of employee is available or not. If available then block employee else display valid message.

Example 1: RishiGupta@gmail.com
Output 1: User successfully blocked from system.
Example 2: RishiBajaj@gmail.com
Output 2: No such user found.

B) Do as directed:
1. What is the use of try-catch block?
2. Write a Java program to generate and handle a custom exception when the fibonacci of a given number is greater than 100.

OR

B) Do as directed:
1. Enlist the subclasses of InputStream class.
2. Write a Java program to print the content of a file using Character Stream.

Q 6 Answer the following in detail. (Any 2) **[10]**

I) Write a program to declare string using StringBuffer class and perform following operations:
Display capacity of declared buffer.
Append “Please letter” in buffer
Insert “consider ” at index 7
Reverse whole string and display
Create string “Believe in yourself. You are braver than you think, more talented than you know, and capable of more than you imagine.” using String class.
Separate tokens by space and by “,” delimiter using string Tokenizer, count all the tokens and display.

II) Discuss various components of JDBC API.

III) Is JDBC Driver and JDBC API both are same? Enlist types of JDBC Driver. Explain any two JDBC Driver.

UKA TARSADIA UNIVERSITY

BCA-Integrated M.Sc. (IT)
Semester :4
030010405 (2012-13) -060010405 (2012-13)
Object Oriented Software Engineering

Date :26/11/2016 **Time :**2:00PM to 5:00PM
Duration : 3 Hours **Max. Marks:**60

Instructions :

1. Attempt all questions.
2. Write each section in a separate answer book.
3. Make suitable assumptions wherever necessary.
4. Draw diagrams/figures whenever necessary.
5. Figures to the right indicate full marks allocated to that question.
6. Follow usual meaning of notations/abbreviations.

SECTION - 1

Q : 1 (A) Answer the following. **[4]**

I) What is the use of operating procedure manuals?

II) Enlist 4 methodologies of Object Oriented.

III) Draw the life cycle of object oriented model.

IV) Give an example that demonstrates “consistent” characteristic of a good requirement.

1 (B) Answer the following in brief. (Any 3) **[6]**

I) State two key differences between macro process and micro process of Booch methodology.

II) Which model emphasizes the reusability of the code and its circle represents different overlapping phases? Give two advantages of the identified model.

III) Write any four guidelines for conducting FAST.

IV) How change control activity is initiated using change request form?

Q : 2 Answer the following. **[10]**

A) Assume that the customer is aware of all the requirements of the software he needs and requires a large amount of supporting documents too. Identify which model should be used by the developer to satisfy the need of the customer. Elaborate the identified model with the help of a diagram

OR

A) Which technique of object oriented methodology is popularly known as Object Modelling Technique (OMT) and was developed in 1990’s. Explain the identified technique in detail.

B) Consider the purchase of gasoline from an electronic gasoline pump.
(a) Prepare a use case diagram for it. Also consider that the customer pays cash for a gas purchase. Add extend relationship to handle the incremental behavior of paying by credit card outside or paying by credit card inside. Add an include relationship to represent the optional purchase of a car wash.
(b) List and explain the relevance of each actor in a sentence.

OR

B) Draw a class diagram for a simple payroll system. Make necessary assumptions and list each one of them to relate to the class diagram.

Q : 3 Answer the following in detail. (Any 2) **[10]**

A) Explain Extreme Programming model in detail. Also state the rules followed in the agile processes.

B) Describe any two requirement elicitation technique in detail.

C) In order to develop good software the requirement gathering should be done precisely. Explain the characteristics that demonstrates good requirement with the help of an example.

SECTION - 2

Q : 4 (A) Answer the following. **[4]**

I) How basic flow and alternative flow differs from each other?

II) Define Interface object.

III) Write any key question that a tester may ask for generating test case from use case.

IV) State the purpose of software testing.

4 B) Answer the following in brief. (Any 3) **[6]**

I) Give two key differences between structured analysis and object oriented analysis.

II) Identify the two basic classes categorized by rational unified process. Draw notations relevant to each of these classes.

III) What is transitions and branching in context to activity diagram?

IV) What is mutation? How mutation score can be calculated?

Q : 5 Answer the following. **[10]**

A) Assume that a customer decides to upgrade her PC and purchase a DVD player. She begins by calling the sales department of the PC vendor and they tell her to talk to customer support. She then calls customer support and they put her on hold while talking to engineer. Finally, customer support tells the customer about several supported DVD options. The customer chooses a DVD and it is shipped by the mail department. The customer receives the DVD, installs it satisfactorily, and then mails her payment to accounting.Consider the above scenario and construct an activity diagram for this process.

OR

A) A company is manufacturing a new product and must coordinate several departments. The product starts out as a raw marketing idea that goes to engineering. Engineering simulates the function of the product and prepares a design. Manufacturing reviews the design and adjusts it to conform to existing machinery. Engineering approves the revisions and customer service then looks at the design – a good design must enable ready repair. Engineering approves the customer service proposals and ensures that the resulting design still meets the target functionality.Construct an activity diagram, for the above process.

B) A simple digital watch has a display and two buttons to set it, the A button and the B button. The watch has two modes of operation, display time and set time. In the display time mode, the watch display hours, minutes, separated by a flashing colon.
The set time mode has two sub modes, set hours and set minutes. The A button selects modes. Each time it is pressed, the mode advances in the sequence: display, set hours, set minutes, display etc. Within the sub modes, the B button advances the hours or minutes once each time it is pressed. Buttons must be released before they can generate another event.
Construct a state diagram of the watch.

OR

B) Consider the following assumptions for deleting a book from library management system:

1. The system requests that administrator/ DEO specify the accession number.
2. The administrator/DEO enters the accession number. The system retrieves and displays the required information.
3. The system prompts the administrator/DEO to confirm the deletion of the book record.
4. The administrator/DEO verifies the deletion.
5. The system deletes the record.

Construct a sequence diagram of basic flow ‘Delete a book’ considering the above conditions.

Q : 6 Answer the following in detail. (Any 2) **[10]**

A) Discuss the five step process used for deriving test case from use cases with the help of an appropriate example demonstrating each step.

B) What does sequence diagram represents? Explain the six different messages that are used in the UML with an appropriate diagram for each one of them.

C) Explain Boundary value analysis and Equivalence class testing in detail with the help of an example.

UKA TARSADIA UNIVERSITY

BCA/Integrated M.Sc. (IT) (Semester: 4)
030010405(2012-13)/060010405(2012-13)
Object Oriented Software Engineering

Date :24/11/2016 **Time :**2:00PM-5:00PM
Max. Marks:60

Instructions :

1. Attempt all questions.
2. Write each section in a separate answer book.
3. Make suitable assumptions wherever necessary.
4. Draw diagrams/figures whenever necessary.
5. Figures to the right indicate full marks allocated to that question.
6. Follow usual meaning of notations/abbreviations.

SECTION - 1

Q 1 A) Answer the following. **[4]**

I) What is function overriding?

II) Consider a student information system. Identify the classes of that system.

III) Define the term class.

IV) In which application build and fix model is used?

Q 1 B) Answer the following in brief. (Any 3) **[6]**

I) Give proper reasons to justify how a software may become outdated.

II) Write the advantages of prototyping model for software development.

III) What do you mean by object oriented approach?GIve a proper example.

IV) “Encapsulation is also known as information hiding concept”. Justify this statement.

Q 2 Answer the following. **[10]**

A) Draw a use case diagram for LIbrary Management System making proper assumptions wherever required.

OR

A) Draw a use case diagram for Railway Reservation System making proper assumptions wherever required.

B) Write a use case description for the following:
1. User can check available tickets and book movie ticket.
2. User can login and post comment on friend’s post on Facebook.

OR

B) Prepare the Initial requirement document for Student Information System.

Q 3 Answer the following in detail. (Any 2) **[10]**

I) On which principle is agile model based?With proper diagram decribe the key features of agile process.

II) Explain in detail the waterfall model.

III) With example show why Consistent, Unambigious and Feasible are characteristics of good requirement.

SECTION - 2

Q 4 A) Answer the following. **[4]**

I) Identify the type of relationship: Timetable and Program Course.

II) Define : Interface Class.

III) What is the importance of generalization relationship in the context of object oriented analysis?

IV) List the possible actors in a web based Leave Management System.

Q 4 B) Answer the following in brief. (Any 3) **[6]**

I) Identify entity, interface and controller classes for **"issue book"** use case of Leave Management System.

II) List out two advantages of Detailed Design.

III) List any two rules of good programming practice.

IV) Show a UML representation for the class "Car".

Q 5 Answer the following. **[10]**

A) Draw a use case descripion form showing both the basic flow and alternate flow for the "Appointment" process of Hospital Management System.

OR

A) Draw a use case description form showing both the basic flow and alternate flow of "Booking" process of Multiplex System.

B) Write short notes on (i) Peer Reviews (ii) Inspections.

OR

B) Descibe in detail checklist with respect to verification of system development.

Q 6 Answer the following in detail. (Any 2) **[10]**

I) What are the levels of testing? In each level of testing which activities are carried out?

II) Draw model which describes relationship between objects of "Order" use case of Pizza Delivery system. Also summarize relationships used in model.

III) Differentiate between verification and validation techniques and also state which activity comes first in software development?